Vijendra Kumar

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Examination	University	Institute	Year	CPI
Post Graduation	IIT Bombay	IIT Bombay	2024-27	8.9
Graduation	JEC	RGPV	2020-24	8.01
Intermediate/+2	HSC	DPS Nigahi	2019-20	9.6
Matriculation	SSC	DPS Nigahi	2018-19	9.3

EXPERIENCE & PROJECTS

- **Ontology-Aware RAG System for Clinical Guidelines** | **KCDH Lab, IIT Bombay**
  In collaboration with Public Health Foundation of India (PHFI)
  - Architected a **multi-agent ontology-aware RAG system** for tobacco cessation guidelines, with specialized agents for concept extraction, ontology-aware semantic chunking, and validation; processed **10+ clinical documents (avg. 200 pages)** and achieved **85%+ semantic coherence** using a **7-factor boundary scoring** algorithm.
  - Implemented a **hybrid retrieval architecture** combining dense vector embeddings with graph-based ontology traversal, along with a **3-component validation pipeline** (completeness, coherence, accuracy); built a **5+ agent orchestration pipeline** enabling relationship-aware query expansion for clinical question answering.

- **Lightweight State-Space Model for Neurovascular Segmentation** | **KCDH Lab, IIT Bombay**
  - Developed a **lightweight Mamba-based state-space model** (**1.68M parameters**, **36× smaller than U-Net**) for temporal intracranial artery segmentation on **DIAS** and **DSCA DSA** datasets, achieving **82.93% mean Dice** (median 84.38%, max 93.61%), matching U-Net baseline performance (82.35%).
  - Demonstrated the effectiveness of **state-space models for variable-length neurovascular sequences** (5–12 frames) at **1024×1024 resolution**, highlighting superior **parameter efficiency** on specialized medical imaging benchmarks.

- **Bhashini Speech-to-Speech Machine Translation (SSMT)** *(Ongoing)*
  *(NLTM-Bhashini, MeitY, Govt. of India | Guide: **Prof. Pushpak Bhattacharyya**, IIT Bombay)*
  - Contributed to model training and evaluation for **English-Hindi, English-Marathi, and Hindi-Marathi** language pairs, sponsored by **MeitY, Government of India**.

R&D Project

- **Bias Detection in LLMs Using RAG** *(Fall 2024)*
  *(R&D 1 Project | Guide: **Prof. Pushpak Bhattacharyya**)*
  - Built a **RAG-based bias detection framework** using **Phi-3-mini** and FAISS retrieval over the SBIC corpus to classify bias type and target social group from input sentences.
  - Fine-tuned Phi-3-mini on 1,000 samples and benchmarked 3 sentence transformers, achieving **60.4% accuracy** on the RedditBias dataset, a 10% gain over the no-RAG baseline.

- **Intersectional Bias Detection Using Retriever-Based Framework** *(Spring 2025)*
  *(R&D 2 Project | Guide: **Prof. Pushpak Bhattacharyya**)*
  - Proposed a novel **retriever-based framework (BiasRetriever)** for intersectional bias detection using biased/unbiased triplets, and introduced two paragraph-level datasets: **Indic-Intersect** and **Western-Intersect** (7,404 paragraphs total).
  - Scraped and curated **REDDIT_BIAS**, a real-world social media dataset for intersectional bias evaluation; benchmarked on existing **HESEIA** & **REDDIT_BIAS**, outperforming **SOTA LLMs** with strong cross-domain generalization.

ACADEMIC PROJECTS

- **Multi-Hop Question-Answering & Reasoning** *(Fall 2024)*
  *(CS 626: Speech and Natural Language Processing and the Web | Instructor: **Prof. Pushpak Bhattacharyya**)*
  - Implemented **self-prompted chain-of-thought** with **6 reasoning paths** and studied extensibility to **Indic** tasks.

- Infusing **LightRAG** with GPT-4o, increasing EM by **2%** on HotpotQA and 2Wiki, compared to Zero-Shot CoT.
- Experimented **NLP** techniques and CoT models on Indic languages through **iNLTK** for dataset pre-processing.

- **Medical Image Classification Using Vision Models** *(Fall 2024)*
  *(CS 725: Foundations of Machine Learning | Instructor: **Prof. Preethi Jyothi**)*
  - Implemented an AutoML optimized **Vision-Transformer** pipeline for **multi-modal** medical image classification.
  - Exhibited **6.7% average gain** in Accuracy and AUC on multiple **MedMNIST** datasets to **ResNet** and **DenseNet**.

- **AQI Guardian – Real-Time Air Quality Platform** *(Fall 2024)*
  *(CS 699: Software Lab | Instructor: **Om P. Damani**)*
  - Engineered a scalable **Flask** backend architecture integrated with external weather APIs to fetch, parse, and serve real-time pollutant metrics for global metro cities.
  - Developed an automated **BeautifulSoup** web scraping module to parse dynamic e-commerce pipelines, dynamically mapping local pollution indexes to relevant health protective equipment.
  - Mitigated critical pipeline infrastructure challenges including anti-scraping CAPTCHAs, rate-limiting thresholds, and IP-blocking by deploying custom request-throttling structures.

- **Subset Selection for Fine-Tuning LLMs: Utility-Diversity Balanced Approach** *(Spring 2025)*
  *(CS769 – Optimization in ML | Guide: **Prof. Ganesh Ramakrishnan**)*
  - Proposed a **utility-diversity subset selection framework** for LLM fine-tuning, introducing a novel **Chain-of-Thought (CoT) metric** combining confidence divergence, backtrack frequency, and symbolic complexity to score data difficulty.
  - Applied **submodular greedy selection** with SBERT-based diversity scoring on **GSM8K**; fine-tuned **Phi-3** on 600 samples achieving **41% accuracy**, outperforming random and DPP baselines.

- **Region-Grounded Knowledge-Infused BLIP-2 for Medical VQA** *(Fall 2025)*
  *(CS 772: Deep Learning for Natural Language Processing | Instructor: **Prof. Pushpak Bhattacharyya**)*
  - Redesigned the standard **BLIP-2 Q-Former** framework by engineering a novel **Anatomy-Aware Q-Former** that integrates frozen text/vision encoders with clinical priors.
  - Developed a **Multi-Head Anatomy Attention** layer and a learnable **Gated Fusion Unit** to eliminate multi-modal object hallucinations by grounding visual queries in clinical spatial semantics.

PUBLICATIONS & PRE-PRINTS

- **Subset Selection for Fine-Tuning: A Utility-Diversity Balanced Approach for Mathematical Domain Adaptation** *(Pre-print)*
  *Kotecha M, **Vaishya VK**, Gautam S, Racha S.* *arXiv:2505.01523*

SELF PROJECTS

- **Safarnama** storiesbysafarnama.com
  - Founded and built **Safarnama**, a consumer storytelling platform for creating personalized shared story timelines and memory experiences.
  - Led end-to-end **product development, technical architecture, deployment, UX design**, and secure **payment integration** for customer transactions.

COURSES TAKEN

- Foundations of Machine Learning
- Optimization in Machine Learning
- Speech and Natural Language Processing and the Web
- Advanced Machine Learning
- Deep Learning for Natural Language Processing
- Deep Learning for Sequences, Graphs, and Language Models
- Software Lab

POSITIONS OF RESPONSIBILITY

- **Student Companion** | **ISCP, IIT Bombay**
  - Mentored 5 students throughout the year in their academics, social, and extracurricular activities, ensuring their holistic well-being.

TECHNICAL SKILLS

- **General Purpose Programming and Scripting Languages**: C, C++, Python, Bash Scripting
- **Tools, Libraries, and Frameworks:** PyTorch, Numpy, Hugging Face, OpenCV, CUDA, FAISS, cvxpy, Nginx
- **Core Competencies and Domains:** AI, ML, DL, Fine-tuning, Optimizations, GPUs
- **Database Management and Miscellaneous**: MySQL, Docker, Postman, Git, HTML, CSS, LaTeX

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Secured **AIR 515** in **GATE - Computer Science** and **AIR 568** in **GATE - Data Science & AI**. *(2024)*
- Qualified **JEE Advanced**.
- Secured **3rd District Rank** in Class 12th Board Examinations.
- Solved over **500+ problems** on **LeetCode**.
- Scored **100%** marks in Mathemathics in the **CBSE** Board Examinations. *(2018)*